

# Project : RetailTransactionAnalysis

## SQL-Analysis

```
1 • create Database RetailAnalysis;
2
3 • Use RetailAnalysis;
4
5 • CREATE TABLE RetailTransactions (
6     TransactionID INT PRIMARY KEY,
7     Date DATE,
8     ProductName VARCHAR(100),
9     Category VARCHAR(50),
10    Region VARCHAR(20),
11    SalesChannel VARCHAR(20),
12    Quantity INT,
13    UnitPrice DECIMAL(10,2),
14    TotalAmount DECIMAL(10,2),
15    PaymentMode VARCHAR(30),
16    CustomerID INT
17 );
```

```
Select * From RetailTransactions;
```

| Result Grid   Filter Rows:   Edit:   Export/Import:   Wrap Cell Content: |               |            |             |             |        |              |          |           |             |             |            |
|--------------------------------------------------------------------------|---------------|------------|-------------|-------------|--------|--------------|----------|-----------|-------------|-------------|------------|
|                                                                          | TransactionID | Date       | ProductName | Category    | Region | SalesChannel | Quantity | UnitPrice | TotalAmount | PaymentMode | CustomerID |
| ▶                                                                        | 1             | 2025-02-15 | Shoes       | Fashion     | West   | Online       | 4        | 3491.00   | 13964.00    | Cash        | 1036       |
|                                                                          | 2             | 2025-03-22 | Bag         | Accessories | South  | Offline      | 3        | 3080.00   | 9240.00     | UPI         | 1018       |
|                                                                          | 3             | 2025-01-19 | Mobile      | Electronics | South  | Online       | 5        | 35027.00  | 175135.00   | Cash        | 1006       |
|                                                                          | 4             | 2025-04-10 | Jacket      | Fashion     | West   | Offline      | 2        | 5745.00   | 11490.00    | Credit Card | 1030       |
|                                                                          | 5             | 2025-04-07 | Kurta       | Fashion     | East   | Offline      | 2        | 1683.00   | 3366.00     | Net Banking | 1038       |
|                                                                          | 6             | 2025-06-10 | Laptop      | Electronics | South  | Offline      | 3        | 64091.00  | 192273.00   | UPI         | 1028       |
|                                                                          | 7             | 2025-03-01 | Watch       | Accessories | North  | Online       | 2        | 1721.00   | 3442.00     | Cash        | 1036       |
|                                                                          | 8             | 2025-04-22 | Watch       | Accessories | East   | Online       | 3        | 11806.00  | 35418.00    | Cash        | 1008       |
|                                                                          | 9             | 2025-03-14 | Smart Watch | Electronics | East   | Offline      | 2        | 5868.00   | 11736.00    | Net Banking | 1037       |
|                                                                          | 10            | 2025-02-13 | Belt        | Accessories | South  | Online       | 4        | 906.00    | 3624.00     | Credit Card | 1039       |
|                                                                          | 11            | 2025-02-11 | Bed         | Furniture   | South  | Online       | 2        | 26263.00  | 52526.00    | Cash        | 1012       |
|                                                                          | 12            | 2025-03-04 | Bed         | Furniture   | West   | Offline      | 1        | 32189.00  | 32189.00    | UPI         | 1021       |
|                                                                          | 13            | 2025-03-10 | Jeans       | Fashion     | North  | Offline      | 5        | 3743.00   | 18715.00    | Cash        | 1040       |
|                                                                          | 14            | 2025-02-23 | Kurta       | Fashion     | North  | Online       | 4        | 2293.00   | 9172.00     | Net Banking | 1007       |

```
-- 1. Total sales amount per region for the last quarter
SELECT
    Region,
    SUM(TotalAmount) AS TotalSales
FROM RetailTransactions
GROUP BY Region
ORDER BY TotalSales DESC;
```

| Result Grid |        |            | Filter Rows: |
|-------------|--------|------------|--------------|
|             | Region | TotalSales |              |
| ▶           | West   | 3063992.00 |              |
|             | East   | 2734762.00 |              |
|             | South  | 2594442.00 |              |
|             | North  | 2420137.00 |              |

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```
-- 2. Top 5 best-selling products (by revenue)
SELECT
    ProductName,
    SUM(TotalAmount) AS Revenue
FROM RetailTransactions
GROUP BY ProductName
ORDER BY Revenue DESC
LIMIT 5;
```

| Result Grid |             |            | Filter Rows: |  |
|-------------|-------------|------------|--------------|--|
|             | ProductName | Revenue    |              |  |
| ▶           | Laptop      | 2085006.00 |              |  |
|             | Camera      | 1797993.00 |              |  |
|             | Bed         | 1227189.00 |              |  |
|             | Cupboard    | 1099026.00 |              |  |
|             | Mobile      | 924504.00  |              |  |

-- 3. Monthly sales trend across all regions

SELECT

    MONTH(Date) AS MonthNo,

    MONTHNAME(Date) AS MonthName,

    SUM(TotalAmount) AS MonthlySales

FROM RetailTransactions

GROUP BY MonthNo, MonthName

ORDER BY MonthNo;

| Result Grid |         |           |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|-----------|--------------|--------------|---------|--------------------|
|             | MonthNo | MonthName | MonthlySales |              |         |                    |
| ▶           | 1       | January   | 1018497.00   |              |         |                    |
|             | 2       | February  | 1428586.00   |              |         |                    |
|             | 3       | March     | 2005792.00   |              |         |                    |
|             | 4       | April     | 1884350.00   |              |         |                    |
|             | 5       | May       | 2255738.00   |              |         |                    |
|             | 6       | June      | 2220370.00   |              |         |                    |

```
-- 4. Region-wise contribution to total sales (as a %)
SELECT
    Region,
    ROUND(
        SUM(TotalAmount) * 100 /
        (SELECT SUM(TotalAmount) FROM RetailTransactions),
        2) AS ContributionPercentage
FROM RetailTransactions
GROUP BY Region;
```

| Result Grid |        |                        | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------|------------------------|--------------|---------|--------------------|
|             | Region | ContributionPercentage |              |         |                    |
| ▶           | West   | 28.34                  |              |         |                    |
|             | South  | 23.99                  |              |         |                    |
|             | East   | 25.29                  |              |         |                    |
|             | North  | 22.38                  |              |         |                    |

```
-- 5. Compare Online vs Offline sales across all months
SELECT
    SalesChannel,
    MONTHNAME(Date) AS MonthName,
    SUM(TotalAmount) AS Sales
FROM RetailTransactions
GROUP BY SalesChannel, MonthName;
```

|   | SalesChannel | MonthName | Sales      |
|---|--------------|-----------|------------|
| ▶ | Online       | February  | 1009326.00 |
|   | Offline      | March     | 772151.00  |
|   | Online       | January   | 399012.00  |
|   | Offline      | April     | 831419.00  |
|   | Offline      | June      | 1724655.00 |
|   | Online       | March     | 1233641.00 |
|   | Online       | April     | 1052931.00 |
|   | Offline      | February  | 419260.00  |
|   | Offline      | May       | 851368.00  |
|   | Offline      | January   | 619485.00  |
|   | Online       | May       | 1404370.00 |
|   | Online       | June      | 495715.00  |

-- 6. Sales trend by Category

SELECT

Category,

SUM(TotalAmount) AS TotalSales

FROM RetailTransactions

GROUP BY Category

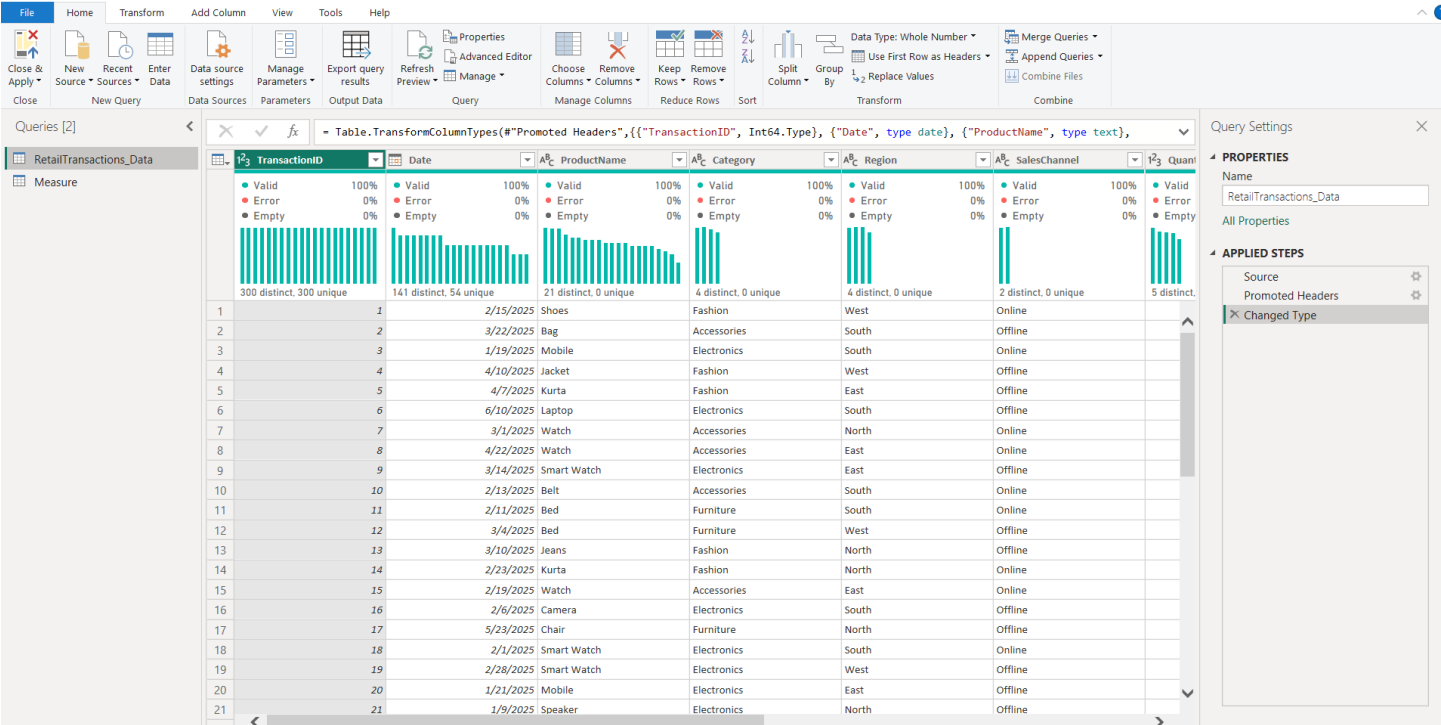
ORDER BY TotalSales DESC;

| Result Grid |             |            | Filter Rows: | Export: | Wrap |
|-------------|-------------|------------|--------------|---------|------|
|             | Category    | TotalSales |              |         |      |
| ▶           | Electronics | 5534635.00 |              |         |      |
|             | Furniture   | 3914764.00 |              |         |      |
|             | Accessories | 793390.00  |              |         |      |
|             | Fashion     | 570544.00  |              |         |      |

```
-- 7. List customers who purchased more than 9 times
SELECT
    CustomerID,
    COUNT(*) AS PurchaseCount
FROM RetailTransactions
GROUP BY CustomerID
HAVING COUNT(*) > 9;
```

| Result Grid |            | Filter Rows:  | Export: | Wrap Cell |
|-------------|------------|---------------|---------|-----------|
|             | CustomerID | PurchaseCount |         |           |
| ▶           | 1007       | 10            |         |           |
|             | 1029       | 10            |         |           |
|             | 1042       | 10            |         |           |
|             | 1003       | 10            |         |           |

# Power BI Dashboard



Total Sales = `SUM(RetailTransactions_Data[TotalAmount])`

Total Transactions = `COUNT(RetailTransactions_Data[TransactionID])`

Unique Customers = `DISTINCTCOUNT(RetailTransactions_Data[CustomerID])`

Average Order Value =  
`DIVIDE(  
    [Total Sales],  
    [Total Transactions]  
)`

# Retail Sales Performance Dashboard

Total Sales  
10.81M

Total Transactions  
300

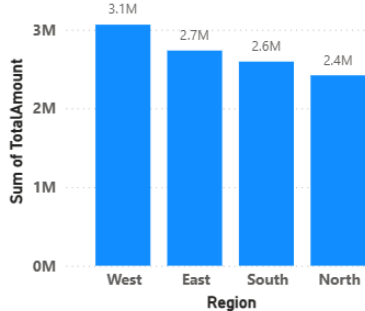
Unique Customers  
50

Average Order Value  
36.04K

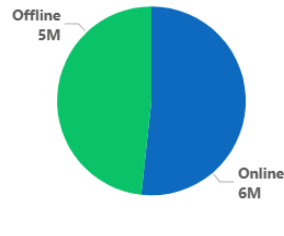
## Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

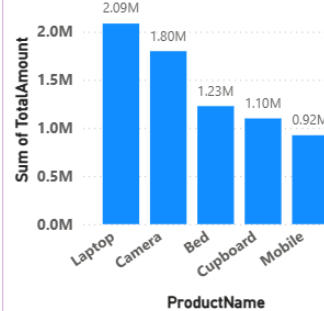
## Region-wise Sales Performance



## Sales Channel Distribution



## Top 5 Best-Selling Products



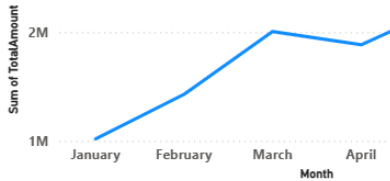
## Category

- ☐ Accessories
- ☐ Electronics
- ☐ Fashion
- ☐ Furniture

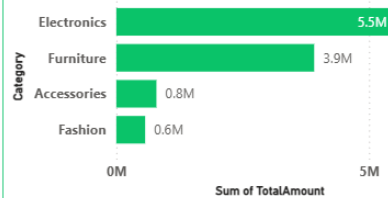
## Sales Channel

- ☐ Offline
- ☐ Online

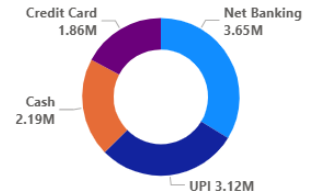
## Monthly Sales Trend Analysis



## Category-wise Revenue Analysis



## Payment Method Distribution



# Retail Sales Performance Dashboard



Total Sales  
2.73M

Total Transactions  
77

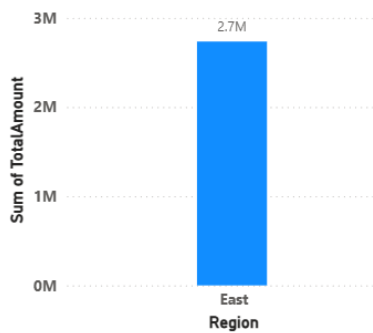
Unique Customers  
40

Average Order Value  
35.52K

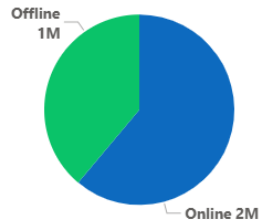
## Region

- ☒ East
- ☐ North
- ☐ South
- ☐ West

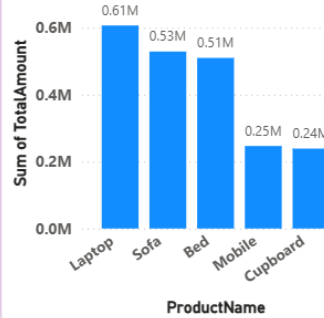
## Region-wise Sales Performance



## Sales Channel Distribution



## Top 5 Best-Selling Products



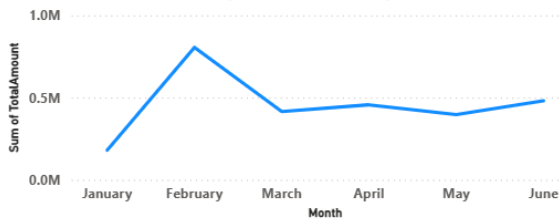
## Category

- ☐ Accessories
- ☐ Electronics
- ☐ Fashion
- ☐ Furniture

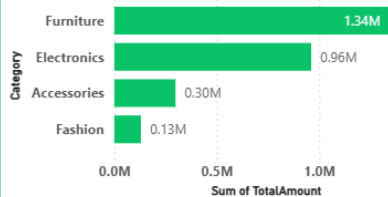
## Sales Channel

- ☐ Offline
- ☐ Online

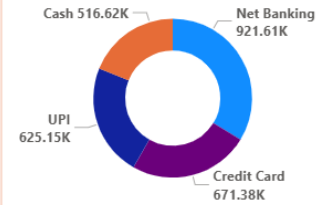
## Monthly Sales Trend Analysis



## Category-wise Revenue Analysis



## Payment Method Distribution





# Retail Sales Performance Dashboard

Total Sales  
129.94K

Total Transactions  
20

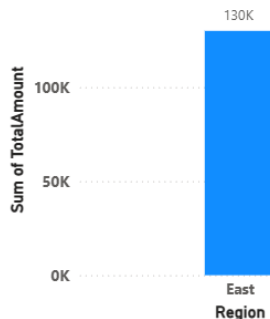
Unique Customers  
17

Average Order Value  
6.50K

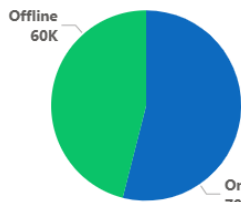
## Region

- ☒ East  
☐ North  
☐ South  
☐ West

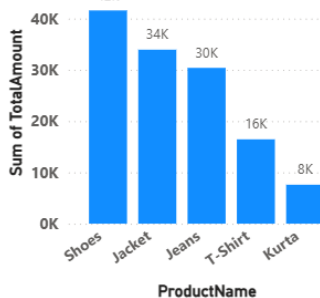
## Region-wise Sales Performance



## Sales Channel Distribution



## Top 5 Best-Selling Products



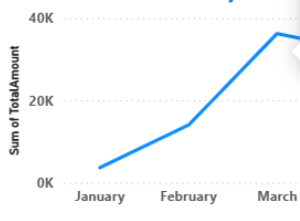
## Category

- ☐ Accessories  
☐ Electronics  
☒ Fashion  
☐ Furniture

## Sales Channel

- ☐ Offline  
☐ Online

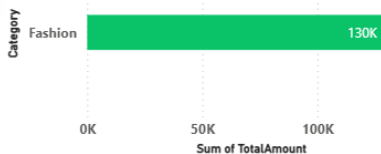
## Monthly Sales



March

Sum of TotalAmount 36270

## Category-wise Revenue Analysis



## Payment Method Distribution

